

DMX6180

Research Methodology and Project Identification

**Enhancing Accuracy in Automated Solid Waste
Segregation**



By

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Declaration

“I hereby declare that this submission is my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of a university or other institute of higher education, except where due acknowledgment is made in the text.”

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Acknowledgement

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Abbreviations

- MSW - Municipal Solid Waste
- WM - Waste Management

Chapter 01: Introduction

Rapid population growth and urbanization have significantly increased the quantity and complexity of municipal solid waste (MSW) generation (Cheng, 2020) (World Bank, 2022). This trend poses major challenges for effective waste management.

Recycling

While recycling organic materials is relatively easier due to composting methods, dealing with other forms of solid waste presents a more complex scenario. Organic materials decompose quickly with the help of microorganisms, while inorganic materials break down slowly as they are not affected by these organisms (Ayilara, 2020). Organic waste from living things returns to the environment faster than inorganic waste.

Non-organic waste, such as plastics, metals, glass, and electronic waste, requires specific and often sophisticated recycling processes. Recycling contamination is a major problem in the recycling industry (CLEANAWAY, 2023). Contamination occurs when materials are sorted into the wrong recycling category or when a recycler is given something that is difficult or impossible to recycle.

The separation and processing of these materials are crucial for efficient recycling. Each type of non-organic waste has its own recycling pathway. For example, plastics need to be sorted by type, as different plastics have different melting points and chemical compositions. Similarly, electronic waste requires careful handling due to its hazardous components, like lead and mercury (Ankit, 2021).

Even recoverable materials, like plastics and paper products, can act as contaminants if they are placed in the wrong recycling container. Contamination can cause major delays and can also cause entire loads of other materials to be contaminated and landfilled. Contaminated paper cannot be processed with clean paper as the paper fibers will not separate from the oils at the recycling facilities.

Environmental and Health Problems

Poor disposal methods, such as open dumping and land filling, worsen environmental pollution. The contamination of soil, water, and air as a result of these practices poses severe risks to public health and ecosystems. For instance, leachate from landfills can get into groundwater, leading to waterborne diseases (typhoid, cholera, and infectious dysentery) with the time (El

Mouine, 2022). Similarly, the decomposition of organic waste in landfills releases methane, a potent greenhouse gas contributing to climate change.

Public Awareness and Participation

Public awareness and participation are key components in enhancing the accuracy of solid waste segregation. Educating the public about the importance of segregation, the correct ways to dispose of different types of waste, and the impact of contamination is essential. However, it's difficult to ask the people to separate waste into more than 4 categories.

Current Situation in Sri Lanka

Sri Lanka is facing considerable challenges in managing its solid waste, exacerbated by the high volume and specific nature of the waste produced. The country generates around 7,000 tons of solid waste each day (UNDP (United Nations Development Programme), 2022), a significant portion of which, approximately 65-66%, is perishable organic material. This situation is particularly acute in the Western Province, including the Colombo region, which is responsible for nearly 60% of the country's total waste generation. On an individual level, the average Sri Lankan generates between 0.4 to 1.0 kilograms of waste daily (Kumara, 2020).

In public areas, separate bins for organic, paper, polythene, and glass wastes are provided, reflecting an awareness of the importance of segregation. However, this initial segregation often does not align with the categories used by the Colombo municipal council, which classifies waste as degradable, recyclable, or non-recyclable (Colombo Municipal Council, 2024).

According to data published by the Waste Management Authority of the Western Province, it generates approximately 3,000 metric tons and collects 1,678.5 metric tons of municipal solid waste (MSW) daily. Out of this, a total of 759 metric tons per month (about 1.7% of the total MSW) is recycled by local authorities. Composting plays a significant role, with 5,539 metric tons being processed monthly, accounting for 12.3% of the total waste management effort. Organic materials are easily and cheaply recycled using composting. Even if the garbage is collected separately as organic, paper, polythene, and glass, the problem does not dissipate. When it comes to non-organic waste, it should be either segregated accurately for recycling or used as energy (Waste Management Authority of Western Province, Sri Lanka, 2023). The breakdown of waste management efforts is as follows:

Table 1: Waste Management in Western Province

Strategy	Monthly Amount (metric tons)	Percentage of Total Management Effort
Recycling	759	1.70%
Composting	5,539	12.30%
Energy Recovery	20,266	45%
Open Dumping	18,342	40.80%

Source: Waste Management Authority of the Western Province (WMA-WP) 2023

Energy recovery from waste constitutes a significant part of the waste management strategy in the Western Province, as shown in table 1, with 20,266 metric tons per month (45%) being directed for this purpose. However, despite these efforts, a considerable amount of waste, around 18,342 metric tons per month (40.8%), is still subjected to open dumping (Waste Management Authority of Western Province, Sri Lanka, 2023).



Figure 1.1: Anuradhapura Manual Waste Segregation Facility

Few facilities for non-organic solid waste segregation exist, like the Anuradhapura Solid Waste Management Facility (Land and Ecosystems (WLE), International Water Management Institute (IWMI), 2020). In the Anuradhapura facility, as shown in the figure 1, a conveyor belt is used, and segregation is done by manual labor. However, considering the amount of waste collected, this is not efficient enough to effect a significant percentage and causes a bottleneck, with the facility becoming a dump site.

Advanced Automated Solid Waste Segregation Systems

Advanced automated solid waste segregation systems have gained popularity in developed countries as a means to streamline the sorting process and reduce manual labor. These systems primarily rely on single method sorting techniques, utilizing advanced technologies such as Magnetic Sensors or Near-Infrared (NIR) Sensors to detect and separate specific types of materials. Magnetic Sensors are particularly effective in separating ferrous metals from the waste stream, while NIR Sensors can identify and sort different types of plastics based on their unique spectral signatures. These technologies enable the systems to quickly and efficiently sort large volumes of waste with minimal human intervention.

However, these systems often focus on detecting specific materials without the capability for rechecking and confirmation, which can lead to a notable gap in accuracy. For example, commercial-level industrial facilities, as shown in figure 2, from Benzer (Benzer, 2024) emphasize detecting specific materials but lack the ability to double-check the sorted waste, potentially resulting in misclassification or contamination. This limitation occurs from the systems' reliance on a single sorting method, which may not account for variations in material composition or the presence of contaminants. As a result, some materials may end up in the wrong sorting streams, reducing the overall efficiency and effectiveness of the recycling process.



Figure 1.2: Municipal Solid Waste Recycling Facility By Benzer

1.1 Aim

1.2 To test the hypothesis that a multi-sensor approach integrating advanced technologies can enhance the accuracy and efficiency of automated solid waste segregation for non-organic waste materials.

1.3 Objectives

1. To quantify current accuracy levels and identify improvement potential using multiple sensors for cross-verification.
2. To analyze limitations of single-method technologies in handling real-world waste complexity and variability.
3. To develop and test a multi-sensor approach for enhanced classification and sorting accuracy.
4. To evaluate the environmental, economic, and social impact of implementing the multi-sensor system.
5. To assess market readiness and scaling potential of the developed solution.

1.4 Problem Statement

Current automated solid waste segregation systems rely on single-method technologies, which have limited accuracy and efficiency in handling the complexity and variability of real-world non-organic waste streams, leading to high levels of contamination, reduced recycling efficiency, and increased environmental and economic costs. Despite advancements in sensor technologies and sorting algorithms, there is a lack of research on the hypothesis that integrating multiple sensors for cross-verification and confirmation can enhance the performance of these systems. This research aims to address this gap by developing and testing a multi-sensor approach that combines advanced technologies to improve the classification and sorting accuracy of non-organic waste materials, such as plastics, metals, glass, and e-waste. By investigating this hypothesis, this study seeks to contribute to the development of more efficient, accurate, and sustainable solid waste management practices..

Chapter 02: Literature Review

Table 2: Literature Review

R.P. No	Title of The Paper	Conference Name, Date, and Author of Publication	Proposed Method & Application	Expanded Adaptive Feature and Achievements	Expanded Research Gap
1	"Artificial Intelligence in Automated Sorting in Trash Recycling" (Costa, 2018)	XV Encontro Nacional de Inteligência Artificial e Computacional (ENIAC 2018), Costa et al., October 22, 2018	Automated garbage sorting system using machine learning models for classifying waste into glass, metal, paper, and plastic.	The system applies SVM and CNN to differentiate waste types. Accuracy: 93% Note: However the testing is done on easy to identifiable images, so the accuracy would be much lower for real waste.	The error margin of 7% highlights the need for manual intervention, indicating a necessity for algorithmic improvements to reduce the rate of misclassification.
2	"Automated Recycling Separation Enabled by Soft Robotic Material Classification" (Chin, 2019)	IEEE International Conference on Soft Robotics, Chin et al., 2019	Soft robotic gripper with integrated sensors for material classification and sorting in recycling.	Developed a cutting-edge puncture-resistant soft robotic gripper equipped with shearing auxetics and capacitive sensors, advancing material classification techniques in recycling. Accuracy: 63% to 85%	The system's lack of robustness presents a barrier to practical application, with material classification under real-world conditions remaining a challenge.
3	"Automatic Sorting of Metal and Non-Metal Objects" (Sharan, n.d.)	[Thesis at KTH, School of Industrial Engineering and Management (ITM)], Sharan and Iskander, 2022	Prototype for automatic sorting of metal and non-metal waste using inductive and ultrasonic sensors.	This prototype showcases the potential of inductive proximity and ultrasonic sensors to effectively segregate metals from non-metals, demonstrating the application-specific utility of these sensor technologies. Accuracy for Metal objects 56% and Non-metal objects 43%	The technology's applicability is confined to sorting only metal and non-metal objects, thereby restricting its use in more complex sorting tasks involving a variety of materials.
4	"VARISORT: Multi-Sensor Sorting Systems for the Recycling Industry" (VARISORT Multi-Sensor Sorting Systems for the Recycling Industry, 2023)	Sesotec GmbH,	Conveyor-based multi-sensor sorting system for recycling streams (plastics, metal, e-waste, solid waste).	The VARISORT system integrates a trio of sensors to sort a range of materials. It includes a color and shape sensor, a metal sensor, and an NIR sensor, marking a significant advancement in sorting technology.	The system's reliance on a single pass detection method without cross-verifying the materials with multiple sensors could lead to sorting inaccuracies, suggesting a need for implementing a more comprehensive verification process.

In the first paper by Costa (2018), the use of machine learning models in automated garbage sorting is explored. While the system shows high accuracy, it's primarily tested on easily

identifiable images, suggesting potential limitations in real-world scenarios. This highlights the need for systems that can handle more complex and varied waste types.

Chin's 2019 study introduces a soft robotic gripper for material classification in recycling. Its innovative approach progressive in material handling, still it faces challenges in strength and real-world applicability.

Sharan's thesis presents a prototype for sorting metals using inductive and ultrasonic sensors. While effective in distinguishing metals, its application is limited, underlining the necessity for more versatile systems capable of handling diverse materials.

Lastly, the VARISORT system described by Sesotec GmbH integrates multiple sensors for sorting various materials. NIR Sensor uses near-infrared technology to identify different types of materials based on their spectral signature. It's particularly effective for sorting plastics, paper, and organic waste. C-Sensor sorts objects based on color and shape, which is useful for separating materials like glass and plastic items that may have specific color-related properties. M-Sensor (The Metal Sensor) is designed to detect metal objects, distinguishing between different types such as ferrous, non-ferrous, and stainless steel, enhancing the sorting of metallic waste from other types.

Even if advanced, it relies on single-method detection, which could lead to inaccuracies, highlighting the potential for multi-sensor approach that can cross-verify materials.

Chapter 03: Methodology

The literature review identified that Multi-Sensor Sorting Systems are highly efficient in waste sorting. However, a notable limitation is their lack of rechecking and confirmation capability. This research will test the hypothesis that adapting rechecking and confirmation using multiple sensors can increase the efficiency of these systems.

The research will follow the steps shown in Table 3, and the block diagram shown in figure 3, it will begin with a survey to understand public opinions on government spending in automated waste segregation and preferences between manual and automated work environments. The study will then define the operational and technological boundaries.

Using case studies to find out the possible technologies to use. The core experimental phase will involve testing these technologies, specifically focusing on the use of multiple sensors for rechecking and confirmation to enhance efficiency and reduce contamination risks.

Based on the collected data, the most viable solution will be selected. This will lead to the design and development of a prototype. Finally, the study will assess the potential for deploying and scaling the chosen technology, considering market readiness and environmental impact.

Table 3: Methodology Steps

Step	Type	Justification for Research Type
Conducting a Survey on Public Opinion	Quantitative	Quantitative methods are suitable for gathering statistically significant data on public perceptions about government spending and preferences for waste segregation environments.
Defining System Boundaries	Qualitative	Qualitative research will provide a deeper understanding of the operational and environmental scope, and the practical implications of these systems.
Finding Multiple Solutions	Qualitative	Qualitative methods, such as case studies, are crucial for exploring a range of potential technologies, providing in-depth insights and contextual understanding.
Testing Selected Technological Solutions	Mixed Methods	A combination of quantitative (for measuring performance) and qualitative (for expert opinions and contextual analysis) methods will offer a comprehensive evaluation of the technologies.
Selecting the Most Viable Solution	Mixed Methods	Mixed methods will enable a balanced decision-making process, combining objective data analysis with qualitative insights from the testing phase.
Designing and Developing a Prototype	Mixed Methods	The design and development process will benefit from both quantitative feedback (efficiency metrics) and qualitative insights (user friendliness)
Testing the Prototype	Mixed Methods	A mix of quantitative (for performance testing) and qualitative (for usability and practical application) methods will ensure a thorough evaluation of the prototype.
Deployment and Scaling Analysis	Mixed Methods	Mixed research is essential for understanding the broader market readiness, scalability challenges, and societal impact of the technology.

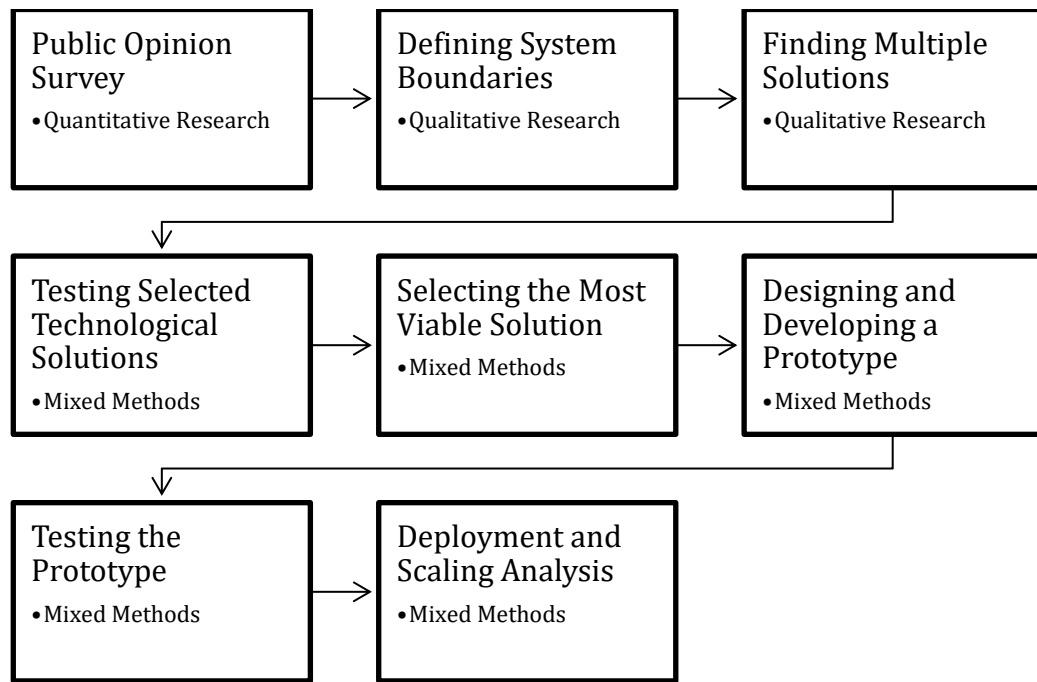


Figure 3.1: Block Diagram for Methodology

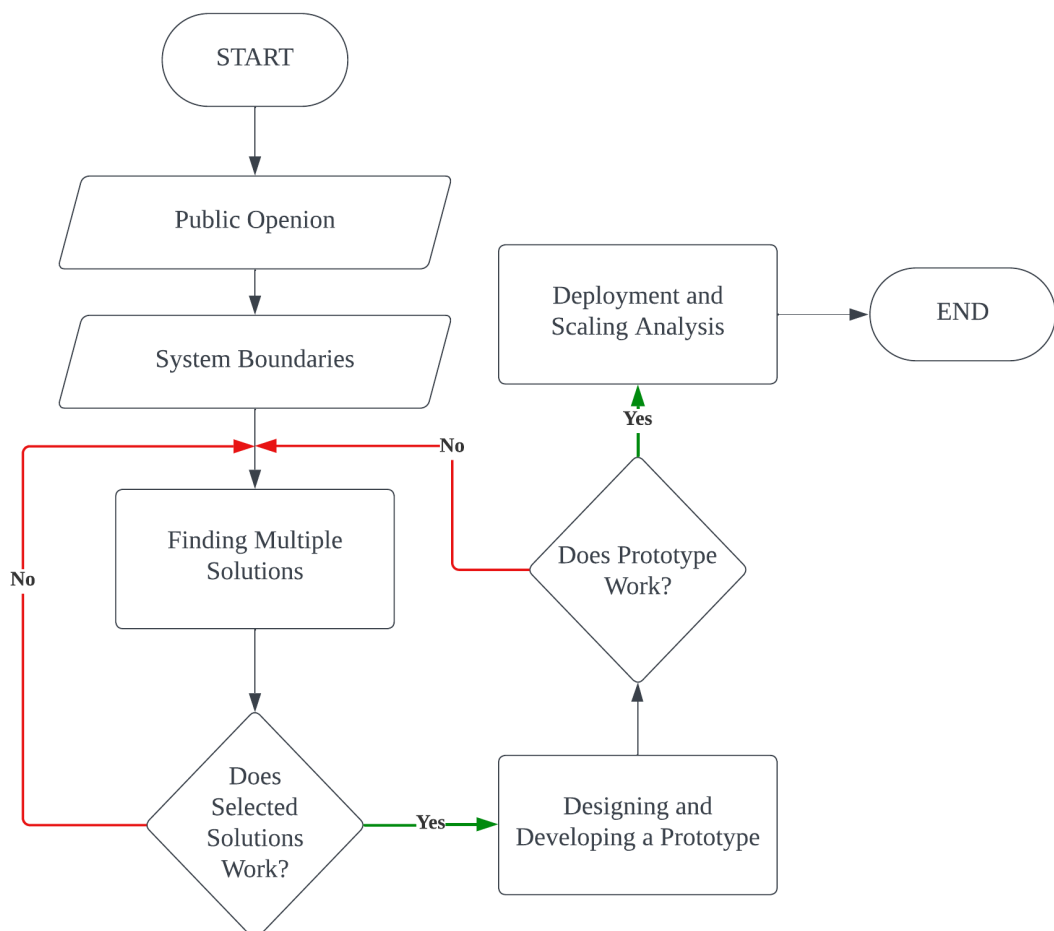


Figure 3.2: Flow Chart of Work Process

Chapter 04: Expected Research Outcome

This research aims to improve non-organic solid waste management. It tests the hypothesis that using multiple sensors, and using cross-checking can enhance the accuracy of automated sorting systems. The goal is to consistently achieve over 95% accuracy in real-world conditions. This involves analyzing current sorting technologies, identifying limitations, and innovating. If successful, this enhancement could redefine waste segregation efficiency standards.

Chapter 05: Sustainability and Ethics

By increasing the accuracy of automated segregation, the initiative contributes to environmental sustainability. This improvement leads to more effective recycling, reducing landfill waste. The economic benefits are there, as the increased recycling efficiency can lead to cost savings. It also generate economic value from waste materials.

Reducing landfill waste contributes to better water quality and overall public health. Importantly, the automation of waste segregation lessens the need for manual handling of waste. This aspect is not only crucial for the physical well-being of workers but also for their psychological health. Our survey indicates a general dislike to working in manual segregation facilities, highlighting the psychological benefits of automated segregation.

References

1. Ankit, S. L. K. V. T. J. S. R. S. S. J. a. B. K., 2021. *Electronic waste and their leachates impact on human health and environment: Global ecological threat and management..* [Online]
Available at: <https://doi.org/10.1016/j.eti.2021.102049>
2. Ayilara, M. O. O. B. O. a. O. O., 2020. *Waste Management through Composting: Challenges and Potentials.* [Online]
Available at: <https://doi.org/10.3390/su12114456>
3. Benzer, 2024. *MUNICIPAL SOLID WASTE RECYCLING EQUIPMENT.* [Online]
Available at: <https://www.bezner.com/products/recycling-solutions/municipal-solid-waste/>

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4. Cheng, J. S. F. Y. J. a. F. H., 2020. *Analysis of the factors that affect the production of municipal solid waste*. [Online]
Available at: <https://doi.org/10.1016/j.jclepro.2020.120808>
 5. Chin, L. L. J. Y. M. K.-B. R. a. R. D., 2019. *Automated Recycling Separation Enabled by Soft Robotic Material Classification..* [Online]
Available at: <https://hdl.handle.net/1721.1/124001>
 6. CLEANAWAY, 2023. *Recycling Behaviours Report 2023*. [Online]
Available at: https://cleanaway2stor.blob.core.windows.net/cleanaway2-blob-container/2023/05/Cleanaway_Recycling-Behaviours-Report-2023.pdf
 7. Colombo Municipal Council, 2024. *GARBAGE COLLECTION DETAILS*. [Online]
Available at: <https://www.colombo.mc.gov.lk/garbage-collection.php>
 8. Costa, B. B. A. P. J. Z. V. P. V. M. G. S. E. S. C. a. S. A. (., 2018. *Artificial Intelligence in Automated Sorting in Trash Recycling*. [Online]
Available at: <https://sol.sbc.org.br/index.php/eniac/article/download/4416/4340>
 9. El Mouine, Y. E. H. A. M. M. V. V. Y. H. a. D. H., 2022. *Groundwater Contamination Due to Landfill Leachate—A Case Study of Tadla Plain*. [Online]
Available at: <https://doi.org/10.3390/environsciproc2022016053>
 10. Kumara, A. a. P. A., 2020. *Household waste disposal mechanisms in Sri Lanka: Nation-wide survey evidence for their trends and determinants*. [Online]
Available at: <https://doi.org/10.1016/j.wasman.2020.06.028>.
 11. Land and Ecosystems (WLE), International Water Management Institute (IWMI), 2020. *Solid and Liquid Waste Management and Resource Recovery in Sri Lanka: A 20 city analysis..* [Online]
Available at: <https://publications.iwmi.org/pdf/H050009.pdf>
 12. Ram Bawankule, V. G. K. I. K. S. a. J. A., 2023. *A Review Paper on Techniques for Collection and Segregation of Solid Waste*. [Online]
Available at: https://doi.org/10.1007/978-981-99-3485-0_53.
 13. Sharan, V. (., n.d. *Automatic waste sorter Automatic sorting of metal and non-metal objects*. [Online]
Available at: <https://kth.diva-portal.org/smash/get/diva2:1708347/FULLTEXT01.pdf>

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14. UNDP (United Nations Development Programme), 2022. *Scaling up Sustainable and Low-Carbon Municipal Solid Waste Management and Resource Recovery*. [Online]
Available at: <https://www.undp.org/srilanka/press-releases/scaling-sustainable-and-low-carbon-municipal-solid-waste-management-and-resource-recovery>
15. VARISORT Multi-Sensor Sorting Systems for the Recycling Industry, 2023. *VARISORT Multi-Sensor Sorting System*. [Online]
Available at:
https://www.sesotec.com/sites/593fc2aac25e5b0640a20ff8/content_entry59e75c78ffe9ecdb05
16. Waste Management Authority of Western Province, Sri Lanka, 2023. *Waste Management in Western Province*. [Online]
Available at: <https://wma.wp.gov.lk/>
17. World Bank, 2022. *Solid Waste Management*. [Online]
Available at: <https://www.worldbank.org/en/topic/urbandevelopment/brief/solid-waste-management>.

Appendix: Survey on SWM for Public Opinion

The survey has done with 44 people to get an rough idea. In this survey, as shown in figure 4, we could conclude that 63.6% people hand over their garbage to garbage collectors. It's good trend that 47.7% percentage use household recycling. But recycling is limited to organic materials. For non-organic materials, they use burning, burying under soil, and put them to water as disposing methods which leads serious environmental concern in all those steps. So, the collection of most of solid, non-organic garbage in both urban and rural areas is important.

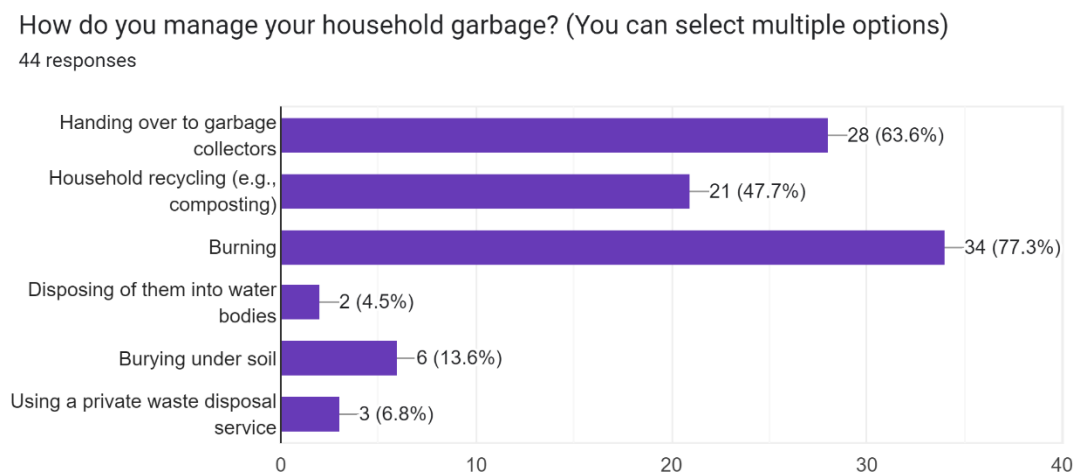


Figure 7.1: How do you manage your household garbage? - Survey Question

And 95.5% of people believes that there should be a percentage allocation from national GDP , as shown in figure 5, for the garbage recycling as a necessary thing to sustain the environment for the future in this industrial era. Even 20.5% of people believe allocating more than 2%, which kind not of practical, but it indicates that the public awareness of recycling is good.

What percentage of the national GDP do you think should be allocated to proper garbage recycling?
44 responses

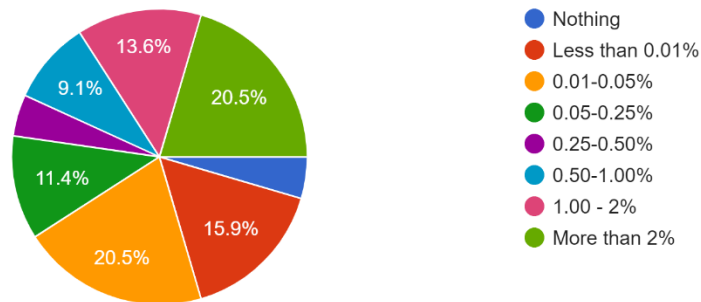


Figure 7.2: What percentage of the national GDP do you think should be allocated to proper garbage recycling?

As shown in figure 6, it appears in most work environments, 81.2% have the access to waste separation facilities. This is good, but 6.8% claims that they don't even use them regularly. Those problems can be fix easily with awareness programs inside the work environment. A significant potion, 15.9% don't have such facilities.

Do you have access to waste separation facilities (e.g., separate bins for different types of waste) at work/ university?
44 responses

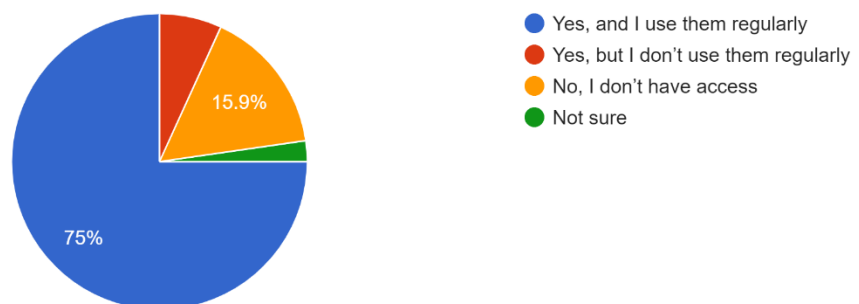


Figure 7.3: Do you have access to waste separation facilities (e.g., separate bins for different types of waste) at work/ university?

When it comes to whether people like or not to work in these environments, I asked two questions, as shown in figure 7.

-
1. Would you consider working in a manual garbage sorting facility where people use hands to sort garbage? (The position will be based on your qualifications)
 2. Would you consider working in a fully automated garbage sorting facility where machines are used to sort garbage? (The position will be based on your qualifications)

Would you consider working in a manual garbage sorting facility where people use hands to sort garbage? (The position will be based on your qualification)

44 responses

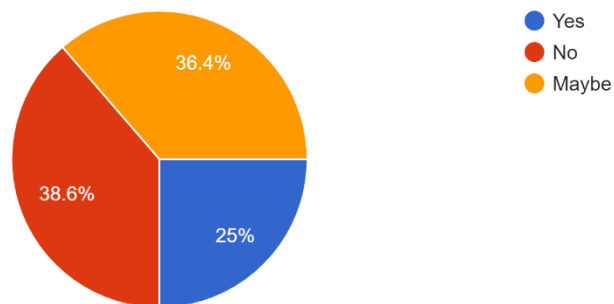


Figure 7.4: Would you consider working in a manual garbage sorting facility where people use hands to sort garbage?

Based on the survey, it's clear that the majority of people don't like to work in manual garbage sorting facilities. But if it's a fully automated one, people willing work there, as shown in figure 8.

Would you consider working in a fully automated garbage sorting facility where machines are used to sort garbage? (The position will be based on your qualification)

43 responses

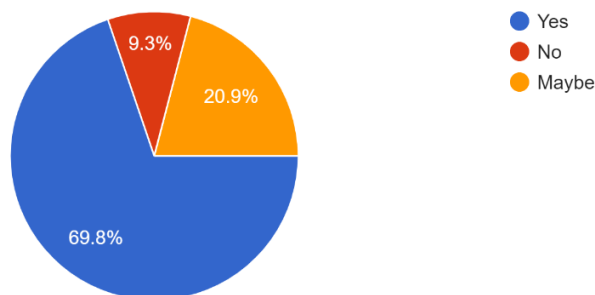


Figure 7.5: Would you consider working in a fully automated garbage sorting facility where machines are used to sort garbage?

The most important thing for my research idea to work is having clean dry non-organic waste separately. Solid waste is collected using several methods. According to our survey, in some areas, waste is collected without sorting, while in others, it is sorted into two categories (organic

and recyclable) before collection. Additionally, in some places, it is even collected as three categories (organic, paper, plastic) or four categories (organic, paper, plastic, glass).

When seeing the data on our survey, as shown in figure 9, it's clear that most people sort their waste when handing over, but there is still 24.1% that doesn't do that. Since we put 85% in dump sites, we can't blame that 24.1% as the cause of this problem. With time, we may try to increase the sorting percentage, but the current collection is enough to start this project.

When you hand over garbage to the garbage collectors, do you sort them by type?

29 responses

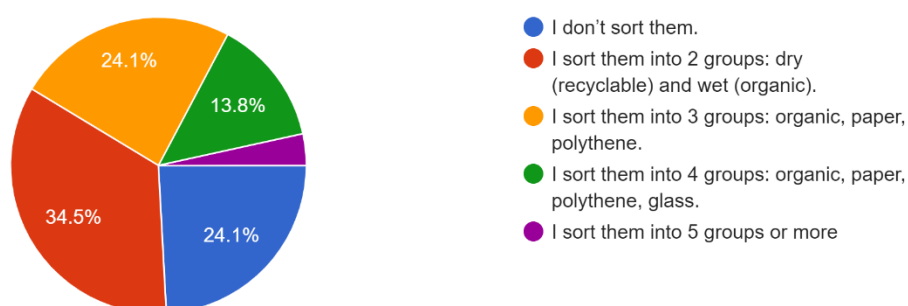


Figure 7.6: When you hand over garbage to the garbage collectors, do you sort them by type?

And majority of people are not satisfied with the recycling policies in their area.

How satisfied are you with the waste management and recycling policies in your area?

31 responses

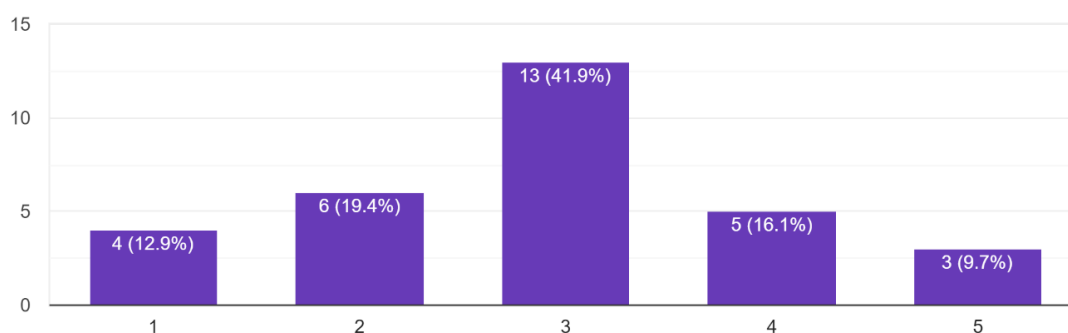


Figure 7.7: How satisfied are you with the waste management and recycling policies in your area?